INCH-POUND

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MILITARY SPECIFICATION

# INDICATOR, ELECTRICAL TACHOMETER 0-110 PERCENT RPM, 2 INCH SIZE, INTEGRALLY LIGHTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two types of hermetically sealed, remote indicating percent speed tachometer indicator, 2 inches in diameter, clamp mounted, with a range of 0-110 percent rpm.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

ERU-5/A Cover glass coated with an antireflection coating MIL-C-675
ERU-5A/A Cover glass coated with multilayer antireflection coating of high efficiency.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of the Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

| | |
|---|---|
| DD-G-451 | Glass, Flat And Corrugated, For Glazing, Mirrors, And Other Uses |
| PPP-B-601 | Boxes, Wood, Cleated Plywood |
| PPP-B-636 | Box shipping, Fiberboard |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEOR, Tinker AFB OK 73145-5990 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods Of |
| MIL-C-675 | Coating Of Glass Optical Elements (Anti-Reflection) |
| DoD-D-1000 | Drawing, Engineering And Associated List |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series: General Specification For |
| MIL-G-9398 | Generator, Tachometer, High Temperature, Two-Pole, Aircraft, Type MB-6 |
| MIL-L-25467 | Lighting, Integral, Instrument, General Specification For |
| MIL-G-26611 | Generator Tachometer GEU/7-A, Miniature |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures And Tables For Inspection By Attributes |
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-130 | Identification Marking Of U. S. Military Property |
| MIL-STD-781 | Reliability Testing For Engineering Development, Qualification, & Production |
| MIL-STD-838 | Lubrication Of Military Equipment |
| MIL-STD-889 | Dissimilar Metals |
| MIL-STD-970 | Standards & Specifications, Order Of Preference For The Selection Of |
| MIL-STD-2073/1 | DoD Materiel Procedures For Development & Application Of Packaging Requirements |
| MS33585 | Pointers, Dial, Standard Design Of Aircraft Instrument |
| MS33639 | Cases, Instrument, Clamp-Mounted, Aircraft |
| MS33678 | Connector, Receptable, Electric, Integral Mounting |

MILITARY

| | |
|---|---|
| MIL-HDBK-781 | Reliability Test Methods, Plans, And Environments For Engineering Development, Qualification, And Production |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).



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American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be address to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3 and 6.4).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-970 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the Military standard parts without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawing.

3.2.2 Standard parts. With the exception in 3.2.1, Military standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

3.3.3 Dissimilar metals. Dissimilar metals are defined by MIL-STD-889 and shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing.



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3.3.4 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent or be protected by hermetic sealing.

3.3.5 Corrosive fumes. The materials as installed in the indicator and under the service conditions specified in 3.5 shall not liberate deleterious fumes.

3.3.6 Protective treatment. When materials are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.4 Design and construction. The indicator shall be designed for operation from the varying-frequency alternating-current output of tachometer generators conforming to MIL-G-9398 or MIL-G-26611. The frequency of the generator output shall be proportional to the speed at which the generator is rotating. The design shall be such that with increasing and decreasing speeds the pointers shall move smoothly at a uniform rate consistent with the speed change. The indicator shall be so constructed that no parts will work loose in service. The indicator shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The indicator shall be so constructed that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases with tools and equipment normally available commercially.

3.5 Performance. The indicator shall not have a Mean Time Between Failure (MTBF) of less than 2500 hours and shall be capable of withstanding the tests specified in Section 4 when subjected to the following conditions:

(a) Indicator starting - Starting and synchronous operation at an indicated speed of 2.5 percent revolutions per minute (rpm) or greater with the following maximum power applied:

(1) Room temperature - 0.525V at 150 cycles per minute (cpm).

(2) -54° C temperature - 0.60V at 150 cpm.

(b) Scale error - Scale error operation at room temperature, -54° C, -35° C, -5° C, and +71° C temperatures in accordance with table I and table II.

(c) Friction error - Friction error not to exceed 1.5 percent at the 5-percent graduation, 0.7 percent at the 20-percent graduation, and 0.5 percent at the 40-, 70-, 85-, and 100-percent graduations.

(d) Position error - Position error not to exceed 0.3 percent when positioned 45 degrees each way from the normal operating position with the indicator operating at 100 percent indication.

(e) Pointer alignment - When the main pointer indicates 90 and 96 percent, the subdial pointer shall indicate 0 and 6 percent, respectively, within 0.20 percent.



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(f) Sealing - Case leakage not to exceed 0.01 micron cubic foot per hour when the filling medium is nitrogen and helium, and 0.1 micron cubic foot per hour when the filling medium is 100 percent helium.

(g) Insulation - Withstand the test voltages specified herein for a period of 5 seconds between connector pins A, B, and C and any metal part of the case or connector shell.

(h) Magnetic effect - Subject the indicator, operating at 90 percent rpm and nonoperating, rotated in increments of 45 degrees from normal, and positioned 5.50 inches from the center of a free-magnet compass having a field intensity of 0.17 to 0.19 oersted, without deflecting the compass more than 3 degrees.

(i) Vibration - Subjected to circular vibration in a plane inclined 45 degrees to the horizontal under the following conditions:

(1) Error - Shall not exceed 0.7 percent rpm when operating at a point between 60 and 90 percent rpm and at an amplitude of 0.009 to 0.011 inch while the frequency is varied from 5 to 50 cycles per second (cps).

(2) Failure - No evidence of failure when operated at 50 percent rpm and vibrated for 3 hours at an amplitude of 0.018 to 0.020 inch while the frequency is cycled once each hour during this test between 5 and 50 cps and return.

(j) Indicator current - The current drawn by each of the 3-phase windings shall not exceed 0.3 amp when connected to 21V 4.200 cpm supply.

(k) Temperature - Subjected to temperature ranging from 62° C to +85° C without failure.

(l) Overspeed and reverse operation - Subjected to a generator shaft speed of 5,000 rpm for 5 minutes and then operated in the reverse direction at a generator-shaft speed of 2,500 rpm for 1 minute without exceeding a scale error of 0.5 percent.

(m) Endurance - Subjected to 1000 hours' operation at 90 $\pm$5 percent indication without damage.

(n) Thermal shock - Subjected to 4 cycles of exposure to water baths maintained at a temperature of 85° $\pm$2° C and 5° $\pm$2° C without evidence of leakage when subjected to the sealing test.

(o) Fogging - Subjected to an ambient temperature of 85° $\pm$5° C for a minimum period of 2 hours and within 20 seconds after being removed from the oven immersed in an ice-water bath, maintained at 0° to 7° C, for a period of 3 minutes.

(p) Salt fog - Subjected to salt fog for a period of 50 hours in accordance with Specification paragraph 4.9 without evidence of external corrosion or deterioration affecting subsequent operation.



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TABLE I Calibration scale errors.

| Generator drive shaft speed (rpm) | Correct indicated speed (percent rpm) | Scale error tolerance room temperature (± percent rpm) | Scale error tolerance (± percent rpm) | | | |
|---|---|---|---|---|---|---|
| | | | -35° C | -54°C | +71°C | -5° C |
| 1/ 0 | 0 | 0.5 | 0.6 | 0.6 | 0.5 | 0.6 |
| 1/ 210 | 5 | 0.5 | 0.6 | 0.6 | 0.5 | 0.6 |
| 1/ 420 | 10 | 0.5 | 0.6 | 0.6 | 0.5 | 0.6 |
| 630 | 15 | 0.5 | --- | --- | --- | --- |
| 1/ 840 | 20 | 0.5 | 0.6 | 0.7 | 0.5 | 0.6 |
| 1,260 | 30 | 0.6 | --- | --- | --- | --- |
| 1/1,680 | 40 | 0.8 | 1.1 | 1.2 | 1.0 | 1.1 |
| 2,100 | 50 | 0.8 | --- | --- | --- | --- |
| 1/2,520 | 60 | 0.8 | 1.2 | 1.4 | 1.0 | 1.1 |
| 2,940 | 70 | 0.8 | --- | --- | --- | --- |
| 1/3,150 | 75 | 0.5 | 1.0 | 1.3 | 0.8 | 0.8 |
| 1/3,360 | 80 | 0.5 | 1.0 | 1.4 | 0.8 | 0.8 |
| 1/3,570 | 85 | 0.5 | 1.0 | 1.4 | 0.8 | 0.9 |
| 1/3,780 | 90 | 0.5 | 1.1 | 1.5 | 0.8 | 0.9 |
| 1/3,990 | 95 | 0.5 | 1.1 | 1.5 | 0.8 | 0.9 |
| 4,200 | 100 | 0.5 | 1.1 | 1.5 | 0.8 | 0.9 |
| 1/4,410 | 105 | 0.5 | 1.2 | 1.6 | 0.8 | 1.0 |
| 4,620 | 110 | 0.5 | --- | --- | --- | --- |

1/ Low and high temperature test points.



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TABLE II. Calibration scale errors.

| Speed (rpm) | Correct indication (percent rpm) | Scale error tolerance room temperature (± percent rpm) | Scale error tolerance (± percent rpm) | | | |
|---|---|---|---|---|---|---|
| | | | -35° C | -54° C | +71° C | -5° C |
| 0 | 0 | 0.5 | 0.6 | 0.6 | 0.5 | 0.6 |
| 1/ 200 | 4.76 | 0.5 | 0.6 | 0.6 | 0.5 | 0.6 |
| 1/ 400 | 9.52 | 0.5 | 0.6 | 0.6 | 0.5 | 0.6 |
| 600 | 14.29 | 0.5 | --- | --- | --- | --- |
| 1/ 800 | 19.05 | 0.5 | 0.6 | 0.7 | 0.5 | 0.6 |
| 1,200 | 28.57 | 0.6 | --- | --- | --- | --- |
| 1/ 1,600 | 38.10 | 0.8 | 1.1 | 1.2 | 1.0 | 1.1 |
| 2,000 | 47.62 | 0.8 | --- | --- | --- | --- |
| 1/ 2,400 | 57.14 | 0.8 | 1.2 | 1.4 | 1.0 | 1.1 |
| 2,800 | 66.67 | 0.8 | --- | --- | --- | --- |
| 1/ 3,200 | 76.19 | 0.5 | 1.0 | 1.3 | 0.8 | 0.8 |
| 1/ 3,400 | 80.95 | 0.5 | 1.0 | 1.4 | 0.8 | 0.8 |
| 1/ 3,600 | 85.71 | 0.5 | 1.0 | 1.4 | 0.8 | 0.9 |
| 1/ 3,800 | 90.48 | 0.5 | 1.1 | 1.5 | 0.8 | 0.9 |
| 1/ 4,000 | 95.24 | 0.5 | 1.1 | 1.5 | 0.8 | 0.9 |
| 1/ 4,200 | 100.00 | 0.5 | 1.1 | 1.5 | 0.8 | 0.9 |
| 1/ 4,400 | 104.76 | 0.5 | 1.2 | 1.5 | 0.8 | 0.9 |
| 4,600 | 109.52 | 0.5 | --- | --- | --- | --- |

1/ Low and high temperature test points.



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(q) Lighting test - Subjected to the lighting tests specified herein.

(r) Indicator acceleration - Withstand the acceleration resulting from an increase of indicated speed of 0 to 30 ± 5 percent within a period of 4 seconds.

(s) Response - Accelerate from 50 ± 1.5 percent to 100 ± 1.5 percent in 2 seconds or less.

(t) Internal examination - Subjected to examination internally for corrosion, deterioration, or damage resulting from all tests specified herein.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part numbers.

3.7 Case. The case shall be in accordance with figure 1 of this specification and shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 0.031 inch.

3.7.1 Material. The hermetically sealed case shall be constructed of a suitable material applicable to hermetic sealing, and the material shall be uniform in texture with a smooth external surface.

3.7.2 Finish. The portion of the indicator case exposed to view from the front of the mounting panel shall be finished in durable dull black. The finish applied to the remainder of the case shall be optional, provided that the finish used will in no way prevent compliance with this specification.

3.7.3 Filling medium. Where a hermetically sealed case is used, the filling medium shall be as least 98 percent pure, free of dust particles, and contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium, shall be either 100 percent helium or a mixture of 88 percent to 92 percent nitrogen and the remainder helium as specified by the procuring activity (see 6.2). The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere.

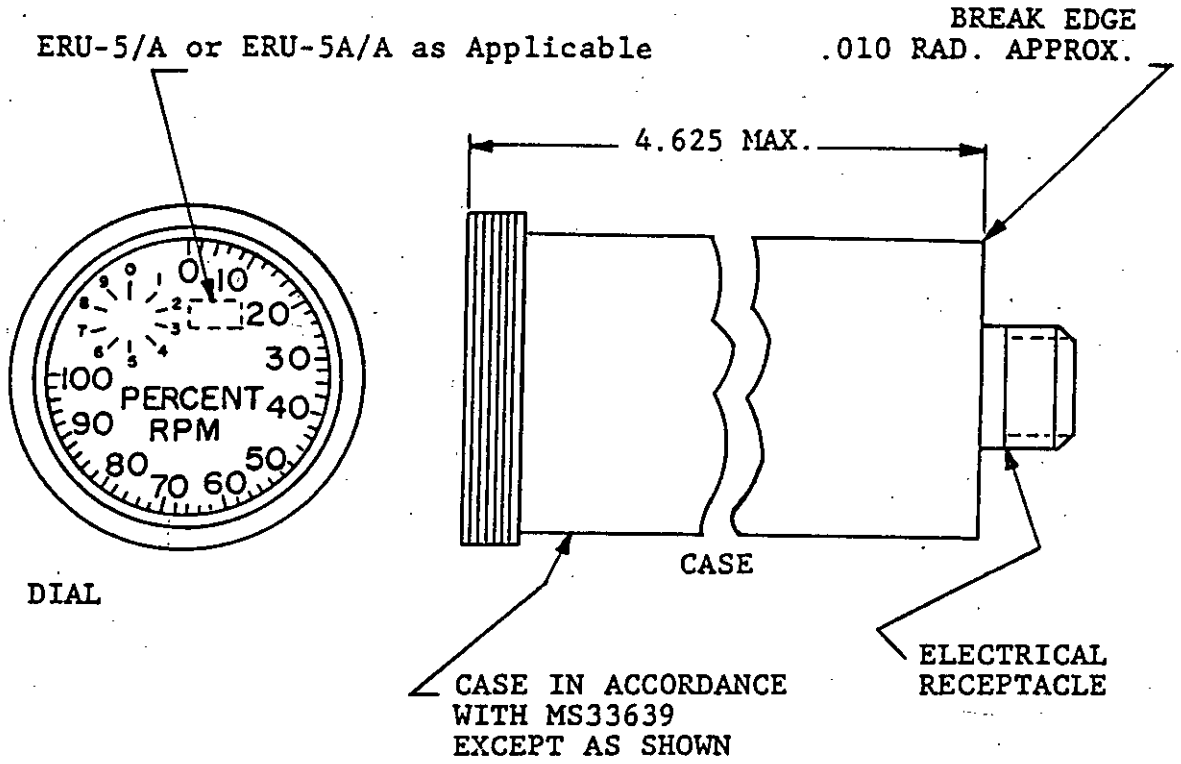
3.7.4 Integral lighting. The indicator shall be integrally lighted in accordance with MIL-L-25467.

3.7.4.1 Lamp location. The lamps shall be contained within the hermetic seal. The lighting system shall be so designed that when lamp replacement is necessary the lamps may be accurately positioned in place. No provision shall be made for lamp replacement in the field. The lamps shall be so installed that easy replacement can be accomplished at a repair depot.

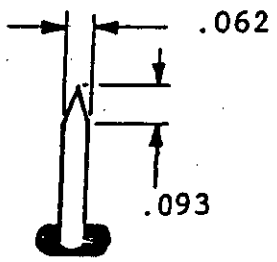
3.8 Cover glass. The cover glass shall be in accordance with type II, AA quality of DD-G-451. The thickness of the glass shall be the minimum



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MAXIMUM DIAMETER OF THE CASE IS EQUAL TO 1.938. DIAMETER FOR MINIMUM DISTANCE OF 1.250 IS EQUAL TO 1.933 +0.005 -0.010. DIMENSION OF 1.250 IS MEASURED FROM FRONT SURFACE OF THE INDICATOR.




SUBDIAL
POINTER

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.016.

FIGURE 1. Case and dial.



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practicable and entirely suitable for the purpose. Any flaws allowed by the federal specification shall not interfere with the readability of the instrument.

3.8.1 Dial to cover glass distance. The distance between the inside surface of the cover glass and the indicator dial shall be the minimum practicable and shall not exceed 0.188 inch.

3.8.2 Reflecting surfaces. All reflecting glass surfaces shall be provided with a reflection-reducing coating which meets the requirements of MIL-C-675 in addition to withstanding the environmental requirements specified herein. For the ERU-5A/A indicator, the following reflectance tolerances shall apply:

Reflectance tolerances

| Angle of incidence | Wavelength millimicrons | Percent reflectance |
|---|---|---|
| 0 degree | 450-675 | 0.6 maximum |
| 0 degree | 425-700 | 0.5 average |
| 30 degrees | 450-625 | 1.0 maximum |
| 30 degrees | 425-700 | 0.5 average |

3.9 Dial. The dial and subdial shall consist of one flat surface with a recess for the subdial pointer. If a recessed disc-type subdial pointer is used, the forward surface of the subdial pointer must be in the same plane as the main dial. The dial material shall be suitable for the purpose and shall be subject to the approval of the procuring activity. The dial shall be securely fastened to the frame of the mechanism in such a a manner that it will no loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be so located that they will not interfere with any dial marking or prevent proper location or dial marking. The dial background and the shaded portion of the pointers shall be black conforming to color No. 37038 of FED-STD-525. The dial shall have a minimum diameter of 1.75 inches when measured across the outside ends of the graduations. The diameter of the subdial, when measured across the outside ends of the graduations, shall be 0.44 $\pm$ 0.016 inch.

3.9.1 Markings. The dial markings shall conform to figure 1 and shall be durable to withstand usage in service. The 100-percent rpm graduation shall be located on the horizontal centerline at the left side of the dial. The dial markings shall be white and conforming to color No. 37875 of FED-STD-595. Unless otherwise specified, the style and proportions of numerals and letters placed on the dial shall conform to Standard MS33558 (see 6.2). Numerals shall distinctly indicate the graduation to which each applies. Any confusion resulting in doubt as to which graduation the numeral applies shall be cause for rejection. The dial graduations shall be uniformly spaced. The dimensions of the markings shall be as specified in table III.



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TABLE III. Dimensions of markings.

| Marking | Height or length +0.016 | Width of line or graduation +0.005 | Color FED-STD-595 |
|---|---|---|---|
| Numerals 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100 | 0.141 | 0.020 | |
| Numerals 1, 2, 3, 4, 5, 6, 7, 8, 9, 0 | 0.094 | 0.016 | |
| 10-percent graduations | 0.141 | 0.025 | |
| 2-percent graduations | 0.093 | 0.016 | No. 37875 |
| Subdial graduations | 0.063 | 0.010 | |
| Lettering "Percent" | 0.093 | 0.016 | |
| Lettering "RPM" | 0.125 | 0.016 | |
| Shaded portion of pointers | ----- | ----- | No. 37038 |
| Lettering ERU-5/A | 0.063 | 0.010 | No. 37038 |

3.10 Electrical receptacle. Each indicator shall be provided with a five-pin electrical receptacle conforming to MS33678-14S-5P located on the vertical and horizontal centerlines of the indicator with the polarizing key of the receptacle at the top. The receptacle shall be of the hermetically sealed type.

3.11 Pointers. The lighted area of the subdial pointer shall conform to figure 1. Its length shall be such that the tip will extend into the scale a distance equal to 0.50 to 0.75 of the length of the graduation. If a recessed disc-type subdial pointer is used, its length shall be such that the outer end of the pointer shall approach the inner ends of the subdial calibration marks within 0.016 inch. The dimensions of the main dial pointer, looking from the front of the indicator, shall conform to MS33585-8. The cross section of the pointers shall be suitable for integral lighting and sufficiently rigid to prevent failure in vibration.

3.12 Indication. The tachometer indicator shall indicate 100-percent speed of the engine when the generator is driven at 4,200 rpm. The indicator shall indicate 0- to 100-percent speed in approximately 270-degree pointer travel. The sub-pointer shall make one complete revolution for each 10-percent change in indication of the main pointer. It shall be possible to read 110 percent by using the subpointer for the last 10 percent, and the main pointer shall indicate at a point equal to 110 percent rpm when the subdial is indicating 110 percent rpm.

3.13 Wiring. The three leads from the indicator windings shall be designated at B, A, and C (conforming to phases 1, 2, and 3). The leads shall be connected to the B, A, and C pins of the electrical receptacle in such a manner that when 3-phase power with phase rotation sequence B-A-C is applied, the resultant indicator reading is positive. The remaining pins, D and E, shall be used for integral lighting.

3.13.1 Electrical connections. Electrical connections for counterclockwise generator drive shaft rotation (when viewed from the shaft end), for operating one or two indicators, shall conform to the wiring diagram shown on figure 2. For clockwise rotation, reverse leads A and B at generator plug. The 5V lighting wiring shall be as shown on figure 2.

3.14 Stops. The indicator mechanism shall be provided with a stop which, upon reverse operation of the indicator motor, shall stop the pointer motion after traveling through an arc equal to not less than 2 percent.



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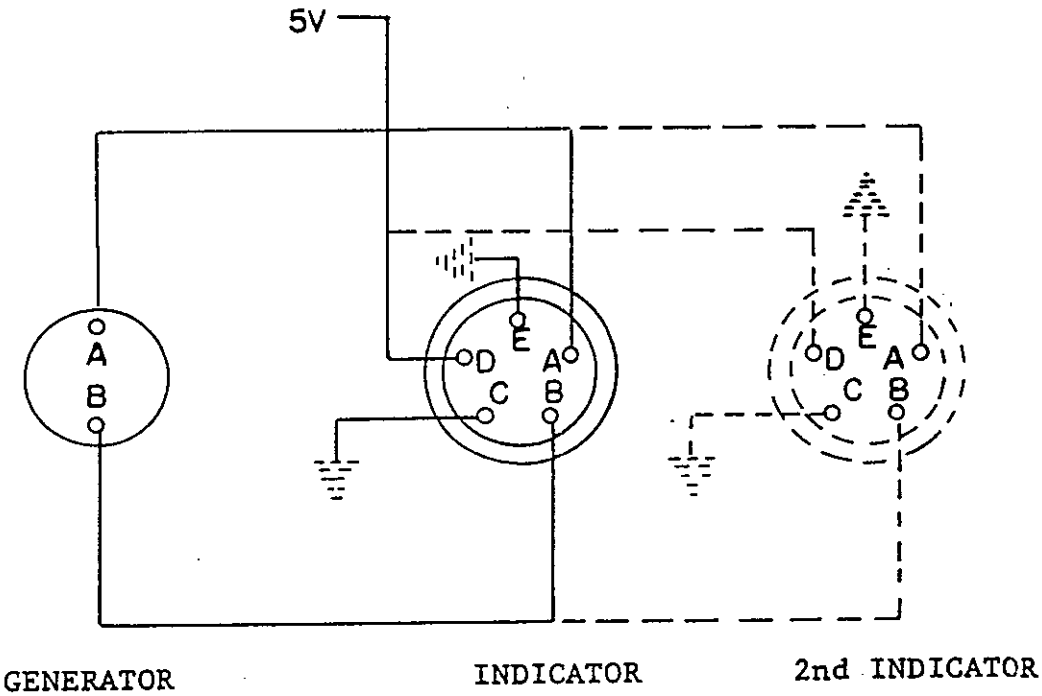



FIGURE 2. Wiring diagram.



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3.15 Lubrication. Lubrication shall be accomplished in accordance with MIL-STD-838, except that the selection of lubricants used shall be approved by the procuring activity.

3.16 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.17 Weight. The weight of each indicator shall not exceed 1 pound.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19 Workmanship

3.19.1 General. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, and painting shall be neat and thorough.

3.19.2 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.19.3 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.19.4 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.19.5 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission



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of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (See 4.3)

b. Quality conformance inspection (see 4.4).

4.3 Qualification inspections.

4.3.1 Qualification inspection. Qualification Inspection shall consist of Reliability Qualification Testing. Seven indicators shall undergo Reliability Qualification Testing (RQT) in accordance with MIL-STD-781 Task 301. Test Plan ID of MIL-HDBK-781 shall be utilized. The Lower Test MTBF ($0_1$) shall be 2500 hours. The combined environments for RQT shall be as specified in paragraph 4.6.

4.4 Acceptance Inspection. Acceptance Inspection shall consist of Production Reliability Acceptance Testing (PRAT). MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes shall be utilized to determine the sample size for Production Reliability Acceptance Testing. PRAT shall be conducted in accordance with MIL-STD-781 Task 302 and utilize Test Plan IVD if MIL-HDBK-781. The Lower Test MTBF ($0_1$) shall be 2500 hours. The combined environments for PRAT shall be as specified in paragraph 4.4.1 and 4.4.2.

4.4.1 Individual tests. Each indicator shall be subjected to the following tests:

| | | |
|---|---|---|
| (a) | Examination of product | (4.6.1) |
| (b) | Room-temperature starting | (4.6.2) |
| (c) | Indicator acceleration | (4.6.3) |
| (d) | Scale error at room temperature | (4.6.4) |
| (e) | Friction error | (4.6.5) |
| (f) | Position error | (4.6.6) |
| (g) | Pointer alignment | (4.6.7) |
| (h) | Sealing | (4.6.8) |
| (i) | Fogging | (4.6.9) |
| (j) | Individual test of MIL-L-25467 | |

4.4.2 Sampling plan A. One indicator selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests:

| | | |
|---|---|---|
| (a) | Individual | (4.4.1) |
| (b) | Low-temperature starting | (4.6.10) |
| (c) | Low-temperature warmup | (4.6.11) |



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(d) Scale error at -54° C (4.6.12)

(e) Scale error at -35° C (4.6.13)

(f) Scale error at -5° C (4.6.14)

(g) Scale error at high temperature (4.6.15)

(h) Insulation (4.6.16)

(i) Magnetic effect (4.6.17)

(j) Vibration error (4.6.18)

(k) Sampling plan A tests of Specification MIL-L-25467

4.4.2.2 Sampling plan B. Unless otherwise specified, 3 indicators selected at random from the first 15 items of the contract or order shall be subjected to the following tests:

(a) Sampling plan A tests (4.4.2.1)

(b) Indicator current (4.6.19)

(c) Low-temperature exposure (4.6.20)

(d) High-temperature exposure (4.6.21)

(e) Vibration failure (4.6.22)

(f) Overspeed and reverse operation (4.6.23)

(g) Indicator endurance (4.6.24)

(h) Thermal shock (4.6.25)

(i) Salt fog (4.6.26)

(j) Sampling plan B tests of Specification MIL-L-25467

(k) Response (4.6.27)

(l) Internal examination (4.6.28)

4.5 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in paragraph 4.

4.5.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditions.



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4.5.2 Tapping. Except where otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.5.3 Vibration. Unless other wise specified, the indicator shall be subjected to circular vibration in a plane inclined 45 degrees to the horizontal plane. The vibration shall be within a frequency range of 5 to 50 cps and at amplitudes within 0.009 to 0.020 inch.

4.5.4 Attitude. Unless otherwise specified, the indicator shall be tested in its normal operating position (dial surface shall be held in vertical plane).

4.5.5 Test voltage. Unless otherwise specified, voltages for indicator operation noted herein are alternating current root-mean-square values. The voltmeter used to measure alternating voltages shall be of a type which is free from frequency errors. Lighting tests shall be made with 5.0 $\pm$ 0.10V direct current applied to the lighting terminals.

4.5.6 Testing. Unless otherwise specified, the following tests shall be conducted on two indicators operation from one tachometer generator.

| | | |
|---|---|---|
| (a) | Room-temperature starting | (4.6.2) |
| (b) | Indicator acceleration | (4.6.3) |
| (c) | Scale error at room temperature | (4.6.4) |
| (d) | Low-temperature starting | (4.6.10) |
| (e) | Low-temperature warmup | (4.6.11) |
| (f) | Scale error at -54° C | (4.6.12) |
| (g) | Scale error at -35° C | (4.6.13) |
| (h) | Scale error at -5° C | (4.6.14) |
| (i) | Scale error at high temperature | (4.6.15) |

4.6 Test methods

4.6.1 Examination of product. Each indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, proper operation, and performance.

4.6.2 Room-temperature starting. This test shall be conducted using the following voltage supplies with frequencies increased from zero cpm up to but not exceeding 150 cpm. The indicator reading taken while the indicator is gently tapped or vibrated shall be not less than 2.5 percent rpm.

4.6.2.1 Variable frequency power supply. One indicator shall be connected to the variable power supply and subjected to the room temperature starting test. The power supply voltage shall be 0.0035 times the frequency in cycles per minute at room temperature. The power supply voltage shall be 0.004 times the frequency in cycles per minute at the low temperature.



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4.6.2.2 Tachometer generator. Two indicators shall be connected to a tachometer generator conforming to MIL-G-9398 or MIL-G-26611 and subjected to the room-temperature starting test.

4.6.3 Indicator acceleration. The indicators shall be connected to a tachometer generator conforming to MIL-G-9398 or MIL-G-26611 as shown in figure 2. The generator speed shall be increased from zero rpm to 1,260 rpm ± 210 rpm within a period of 4 seconds. The indicators shall synchronize with the tachometer generator and their indication shall not exceed the room temperature scale error tolerance specified in table I.

4.6.4 Scale error at room temperature. The indicators shall be connected as shown in figure 2 to a generator conforming to MIL-G-9398 or MIL-G-26611 and tested at points of the scale indicated in either table I or table II. The test shall be made either by driving the generator at a speed to produce the specified indicator readings and reading the speed of the generator or by driving the generator at the specified speeds and reading the indicator. The test shall be made with speeds increasing then with speeds decreasing. With speeds increasing, the speed shall be brought up to but shall not exceed the speed specified to give the desired reading, and with speeds decreasing, the speed shall be brought down to but shall not fall below the speed specified to give the desired reading.

a. The errors at the test points shall not exceed the tolerances specified in the applicable table. When the speed is held constant at any point of the scale, the pointer shall not oscillate over a range greater than 0.5 percent from 0 to 15 percent and 0.3 percent from 15 to 100 percent.

4.6.5 Friction error. The indicator shall be set at 5-, 20-, 40-, 70-, 85-, and 100 percent graduations, or at generator speeds 200, 800, 1,600, 3,200, 3,600, and

4,200 rpm whichever the contractor finds more convenient. After each setting, the indicator shall be tapped and the change in indications noted. This change shall not exceed 1.5 percent at the 5-percent graduation, 0.7 percent at the 20-percent graduation, and 0.5 percent at the 40-, 70-, 85-, and 100 percent graduations.

4.6.6 Position error. The indicator shall be operated at 100 percent with the indicator in the normal (dial vertical) position. The indicator reading shall be noted. The maximum deviation from the normal position reading, when tested in the following positions, shall not exceed 0.3 percent:

(a) Rotated from normal position through 45 degrees to the rear (dial tilted 45 degrees back from vertical position).

(b) Rotated from normal position through 45 degrees to the front (dial tilted 45 degrees forward from vertical position).

(c) Rotated from normal position through 45 degrees to the right (dial vertical).

(d) Rotated from normal position through 45 degrees to the left (dial vertical).

4.6.7 Pointer alignment. The indicator shall be operated so the main pointer will indicate 90 and 96 percent rpm, respectively. The subdial pointer shall then correspondingly indicate 0 and 6, respectively, within 0.20 percent.



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4.6.8 Sealing. The instrument case shall be tested for leaks by means of a mass spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.01 micron cubic foot per hour at a pressure differential of 1 atmosphere. Where 100-percent helium is utilized as the filling medium, the leak rate shall not exceed 0.1 micron cubic foot per hour.

4.6.9 Fogging. The indicator shall be placed in a controlled ambient temperature of 85°F ± 5° for a minimum period of 2 hours. Within 20 seconds after removal from the oven, the indicator shall be immersed to a depth of 0.50 ± 0.25 inch in an ice-water bath maintained at 0° to 7° C. After 3 minutes, the indicator shall be removed from the bath and wiped dry (compressed air shall not be used). The cover glass and dial shall immediately be inspected and any evidence of fog or damage shall be cause for rejection.

4.6.10 Low-temperature starting. Prior to this test, the room temperature scale error test shall be repeated. The indicators shall then be placed in a chamber, the temperature of which is controllable. The temperature of the circulating air shall be reduced to -54° ± 2° C and held at this temperature for the duration of the test. The indicators shall be kept in the chamber for at least 4 hours after the temperature of -54° ± 2° C is reached before the test is started. No voltage shall be applied to the indicator during this period. The indicators, while still at low temperature, shall be subjected to and pass the test specified in 4.6.2.

4.6.11 Low-temperature warmup. Following the starting test at low temperature specified in 4.6.10, the indicator shall be maintained at the -54° C temperature for a period of 2 hours or more without voltage applied. The indicator shall then be instantaneously connected to a tachometer generator operating at 4,200 rpm. The indicator shall read 100 ± 1.5 percent rpm within a period of 15 seconds.

4.6.12 Scale error at -54° C. Following the low temperature warmup test and while still at -54° C, the indicator readings at the points indicated by footnotes in table I or table II shall be determined in the same manner as specified in 4.6.3 with one generator operating two indicators. The errors at the test points shall not exceed the tolerances specified in the applicable.

4.6.13 Scale error at -35° C. The indicator readings shall be determined as specified in the scale error test as -54° C, except that the temperature of the air in the chamber shall be -35° C ± 2° C. The indicators shall be kept at -35° ± 2° C for at least 2 hours prior to the test. The errors at the test points shall not exceed the tolerances specified in the applicable table.

4.6.14 Scale error at -5° C. The indicator readings shall be determined as specified in the scale error test as -54° C, except that the temperature of the air in the chamber shall be -5° C. The indicators shall be kept at -5° ± 2° C for at least 2 hours prior to the test. The errors at the test points shall not exceed the tolerances specified in the applicable table.

4.6.15 Scale error at high temperature. The indicator scale readings shall be determined as specified in the indicator scale error test at -54° C, except that the temperature of the air in the chamber shall be +71° ± 2° C. The indicators shall be kept in the chamber for 4 hours after the temperature of +71° C is reached before the test is started. The errors at the test points shall not exceed the tolerances specified in the applicable table.



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4.6.16 Insulation. To test the insulation, a potential at a commercial frequency shall be applied between pin A, B, or C of the electrical receptacle and any part of the instrument case or the electrical receptacle shell for a minimum period of 5 seconds. For the unsealed instrument, the applied voltage shall be 500V. If the indicator is filled with gas and hermetically sealed, the applied voltages shall be 200V for indicators using the helium filling medium and 400V for the nitrogen-helium mixture filled indicators. There shall be no breakdown of the insulation.

4.6.17 Magnetic effect. The indicator, first not operating and second operating at an indication of approximately 90 percent rpm, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5.50 in inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0 degree, 45 degrees, 90 degrees, 135 degrees, 180 degrees, 225 degrees, 270 degrees, and 315 degrees from the initial positions. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 3 degrees.

4.6.18 Vibration error. The indicator reading shall be determined when the generator is operated at a point between 60 and 90 percent rpm as selected by the Inspector. The indicator shall then be subjected to circular vibration as specified under test conditions at an amplitude of 0.009 to 0.011 inch diameter. The indicator readings shall be determined with the generator operating at the original speed as the vibration frequency is varied from 5 to 50 cps. The readings with vibration shall not vary more than 0.7 percent from the readings without vibration. The amplitude of the pointer oscillation shall not exceed 0.7 percent.

4.6.19 Indicator current. The indicator shall be tested using a 21V, 4,200 cpm supply to determine that the current drawn by any one of the 3-phase windings does not exceed 0.30 amp.

4.6.20 Low-temperature exposure. The room temperature scale error test for increasing speeds only shall be repeated. The indicator shall then be subjected to a temperature of -62° ± 2° C for a period of 48 hours. After the chamber has returned to room temperature and the indicator has remained at room temperature for a minimum period of 4 hours, it shall be subjected to and pass the scale error at room temperature test (4.6.4), increasing speeds only, and the friction test (4.6.5). There shall be no damage which would adversely affect subsequent operation.

4.6.21 High-temperature exposure. The indicator shall be placed within a chamber and the internal temperature of the chamber raised to +71° ± 2° C with an internal relative humidity of not more than 5 percent. The source of heat for the chamber shall be arranged in such a manner that radiant heat shall not fall upon the test specimen. The total volume occupied by a single item of equipment or by several items shall not exceed 50 percent of the internal volume of the test chamber. The indicator shall be maintained at +71° ± 2° C for a period of 24 hours. While still at this temperature, the indicator shall be operated throughout its range, for increasing speeds only, without exhibiting sticking or erratic characteristics. After the chamber has returned to room temperature and the indicator has remained at room temperature for a minimum period of 4 hours, it shall be subjected to and pass the scale error at room temperature test (4.6.4), increasing speeds only, and the friction test (4.6.5). There shall be no damage which would adversely affect subsequent operation.

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4.6.22 Vibration failure. The indicator shall be mounted on the vibration stand in its normal operating position. The test shall be conducted while the indicator is being operated at an indication of approximately 50 percent. The indicator shall be subjected to vibration with a circle diameter between 0.018 and 0.020 inch for a 3-hour period. Once each hour during this test the frequency shall be varied uniformly from 5 to 50 cps and return. After the 3-hour period, the indicator shall be subjected to the scale error at room temperature test (4.6.4), increasing speed only. The readings shall not differ from the readings before vibration by more than 0.5 percent. In addition, no looseness in the mechanism or damage to any part of the instrument shall result from this test.

4.6.23 Overspeed and reverse operation. One or two of the indicator mechanisms shall be electrically connected to a tachometer generator. The reading of the indicators shall be determined when operated at 20 and 90 percent rpm. The indicators shall be at room temperature and the readings shall be taken before the indicators have attained an appreciable temperature rise.

4.6.23.1 Generator and indicators speed. The generator and indicators shall then be operated, first at a generator shaft speed of approximately 5,000 rpm for a period of 5 minutes, and then in the reverse direction of rotation at a generator drive shaft speed of approximately 2,500 rpm for a period of 1 minute.

4.6.23.2 Indicator scale. After the indicators have been allowed to cool to room temperature, the readings of the indicator shall again be determined when operated at 20 and 90 percent rpm. The change in indicated speed at the respective scale points before and after the overspeed test shall be not more than 0.5 percent.

4.6.24 Indicator endurance. The readings of the indicator shall be determined when operated at 20 and 90 percent rpm. The indicator shall be at room temperature and the readings shall be taken before the indicator has attained an appreciable temperature rise. The indicator shall then be operated at an indicator of 90 $\pm$ 5 percent rpm for a period of 1000 hours. The voltage shall be measured at the indicator terminals and shall be 18.5 $\pm$ 1V during the 1000-hour period. At the end of the 1000-hour period and after the indicator has been allowed to cool to room temperature, the readings of the indicator shall again be determined when operated at 20 to 90 percent rpm. The change in reading obtained before and after this test shall be not more than 0.5 percent. No screws or other parts shall become loose or damaged as result of this test. Indicators which have been subjected to all the tests of this specification up to and including this test shall also be subjected to the scale error at room temperature, friction, and position error tests (4.6.4, 4.6.5, and 4.6.6, respectively). Scale error shall not be 0.5 percent more than allowed at room temperature in table I and II. Friction error shall not exceed 1.5 percent at the 5- and 20-percent graduations and 1.0 percent at the 40-, 70-, 85-, and 100-percent graduations. Position error shall not exceed 0.5 percent.

4.6.25 Thermal shock. The indicator shall be subjected to four cycles of exposure to water at 85° $\pm$ 2° C and 5° $\pm$ 2° C without evidence of moisture penetration or damage to coating or enclosures. Each cycle of the test shall consist of immersing the indicator in water at 85° C for a period of 30 minutes and then, within 5 seconds of removal from the bath, the indicator shall be immersed for a period of 30 minutes in the other bath maintained at 5° C. This cycle shall be repeated continuously, one cycle following the other until four cycles have been completed. Following this test, the indicator shall be subjected to the sealing test. No indicator leakage shall occur as a result of this test.



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4.6.26 Salt fog. The indicator shall be subjected to a salt fog test in accordance with Specification paragraph 4.9 for a period of 50 hours. External connections shall be made to the indicator in such a manner as to simulate installed conditions. Following this test, the indicator shall be examined and there shall be no evidence of external corrosion or deterioration which will affect subsequent operation. The indicator shall then be subjected to the insulation and leak tests as specified herein. The external surface of the indicator may be washed with distilled water and air dried prior to conduction of the insulation test.

4.6.27 Response. The test will be conducted using two tachometer generators conforming to MIL-G-9398 or MIL-G-26611. One generator will be operating and stabilized at 50 ± 1.5 percent and the other generator will be operating and stabilized at 100 ± 1.5 percent. The indicator to be tested and the generators will be wired using a switch. This will enable the indicator to be driven by either generator. The indicator to be tested will be operated until it reaches operating temperature (90 percent minimum RPM for 5 minutes). The indicator will then be driven a minimum of 1 minute with the slow (50 ± 1.5 percent) speed generator and allowed to stabilize. The switch will then be thrown thereby switching the indicator to the high (100 ± percent) speed generator and starting a timer at the same time. The timer will be stopped when the indicator reaches an indication of 100 ± 1.5 percent. The time shown on the timer shall not exceed 2 seconds.

4.6.28 Internal examination. The indicator shall be examined internally for evidence of corrosion or deterioration resulting from the tests specified herein. There shall be no evidence or corrosion or deterioration.

4.6.29 Reflectance. The cover glass shall meet the requirements of 3.8.2.

4.7 Rejection and retest. When an item selected from a production run fails to conform to this specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.7.1 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items conform to this specification.

4.7.2 Defects in item already accepted. If the investigation of a test failure indicates defects in items already accepted, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.8 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

4.9 Salt fog test.

4.9.1 Performance record. Prior to conducting any of the tests specified herein, the equipment shall be subjected to a comprehensive operational test under standard conditions and a record made of all data necessary to determine compliance with the applicable equipment specification. These data shall provide the criteria for checking satisfactory performance of the equipment during or after environmental



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tests. Where the applicable performance for the test procedure, a detailed pre-exposure performance record need not be made. Equipment in this latter category shall be operated prior to test to insure that no malfunction exists.

4.9.2 Criteria for failure. Deterioration or change in performance of any components which could in any manner prevent the equipment from meeting functional, maintenance and service requirements during service life shall provide reason to consider the equipment as having failed to comply with the conditions of the test to which it was subjected.

4.9.3 Evaluation of equipment. When so directed in individual test procedures, the equipment undergoing test shall be operated to permit performance data to be obtained, or inspected for evidence of deterioration. The performance data of the equipment under test conditions or following test shall be satisfactorily comparable to that obtained in compliance with 4.9.1. Any deterioration observed shall not exceed that defined in 4.9.2.

4.9.4 Salt fog method.

4.9.4.1 Scope.

4.9.4.1.1 Accelerated-corrosion. This method covers procedures for an accelerated-corrosion test in which specimens are exposed at a temperature of 95°F (35°C) to a fine mist of 5 percent sodium chloride solution. When no solution concentration is specified, the 5 percent solution shall be used. The method constitutes an arbitrary performance test which is useful in the establishment and maintenance of certain standards of quality, although it does not closely simulate any common conditions of service.

4.9.4.1.2 Testing method. The method is applicable to the testing of assemblies, metallic coatings, organic and inorganic coatings on metals, and many nonmetallic materials. The method describes the conditions for conducting the test for specification purposes. It does not prescribe the type of test specimen, or length of exposure periods to be used for any specific product, or the interpretation to be given to the results. It is desirable in the specification of this test, that the significant surfaces for testing, when applicable, be indicated on the drawing, or in the detail specification, contract, or order. The size and other construction details of the apparatus are optional, provided the conditions obtained meet the requirements of the method described herein.

4.9.4.2 Apparatus.

4.9.4.2.1 Apparatus used in the salt fog test shall include the following:

(a) Exposure chamber with racks for supporting specimens

(b) Salt solution reservoir

(c) Means for atomizing salt solution including suitable nozzles and compressed air supply

(d) Means for heating the exposure chamber and controlling its temperature

(e) Means for cleaning the air and humidifying it at a temperature above that of the exposure chamber.



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4.9.4.2.1.1 Exposure chamber. The chamber and all accessories shall be made of material such as glass, hard rubber, plastic, or suitably coated wood which will not affect the corrosiveness of the fog. In addition, all parts that come in contact with test specimens shall be made of materials that will not cause galvanic or electrolytic corrosion. The chamber and accessories shall be constructed and arranged as follows:

(a) There shall be no direct impinging of the spray, or dripping of condensate on the specimens.

(b) The spray shall circulate freely about all specimens to the same degree.

(c) No liquid which has entered the exposure chamber and has come into contact with the test specimens shall return to the salt solution reservoir.

(d) The chamber shall be properly vented.

4.9.4.2.1.2 Atomizer. The atomizer or atomizers used shall be of such design and construction as to produce a finely divided, wet, dense fog.

4.9.4.2.1.3 Air supply. The compressed air entering the atomizers shall be essentially free from all impurities such as oil and dirt. Means shall be provided to humidify and warm the compressed air as required to meet the operating conditions. The air pressure shall be suitable to produce a finely divided dense fog with the atomizer or atomizers used. To avoid any substantial difference between the salt concentration in the fog particles and that of the salt solution before atomization, the atomizing air must have a relative humidity between 95 and 98 percent for the 5 percent solution and of at least 85 percent for the 20 percent solution after release inside the cabinet. Thus, it is necessary to saturate the air at a higher temperature and cool it to 95°F (35°C) in the chamber in order to comply with the requirement for the NaCl content of the collected solution in 4.9.4.4.2.2.1. A satisfactory method is to pass the air in very fine bubbles through a tower containing water maintained at the proper temperature. Temperatures usually found to be satisfactory for various values of air pressure are given the following table:

| Air pressure (p.s.i.) | Temperature | | | |
|---|---|---|---|---|
| | 5 percent solution | | 20 percent solution | |
| | °F | °C | °F | °C |
| 12 | 114 | 46 | 110 | 45 |
| 14 | 117 | 47 | 117 | 47 |
| 16 | 119 | 48 | 119 | 48 |
| 18 | 121 | 49 | 121 | 49 |

4.9.4.2.1.4 Salt solution. The salt used shall be flake salt, substantially free of nickel and copper and containing on the dry basis not more than 0.1 percent of the sodium iodide and not more than 0.3 percent of total impurities. "Butter grade" flake salt has been found to meet these requirements. The 5 percent solution shall be prepared by dissolving 5 ± 1 parts by weight of salt in 95 parts by weight of distilled water, or deionized water containing not more than 200 parts soluble solids per million. The 20 percent solution shall be prepared by dissolving 20 ± 2 parts by weight of salt in 80 parts by weight of distilled, or deionized water



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containing not more than 200 parts soluble solids per million. The solution shall be kept free from solids by filtration or decantation. The solution shall be kept free from solids by filtration or decantation. The solution shall be adjusted to and maintained at a specific gravity of from 1.0268 to 1.0413 for the 5 percent solution and 1.126 to 1.157 for the 20 percent solution when measured at a temperature of 95°F (35°C).

4.9.4.2.1.4.1 Salt solution. The pH of the salt solution shall be so maintained that solution atomized at 95°F (35C°) and collected by the method described in 4.9.4.4.2.2.1 will be in the pH range of 6.5 to 7.2. The pH measurement shall be made immediately after the collected sample is removed from the exposure chamber. The pH shall be measured electrometrically using a glass electrode with a saturated potassium chloride bridge; or colorimetrically using bromthymol blue as indicator. Only diluted c.p. hydrochloric acid of c.p. sodium hydroxide shall be used to adjust the pH.

4.9.4.2.1.4.2 Temperature. When the pH of a salt solution is adjusted at room temperature and atomized at 95°F (35°C) the pH of the collected solution will be higher than that of the original solution due to the loss of carbon dioxide at the higher temperature. The following three methods have been found satisfactory for adjusting the pH of the original solution so that the pH of a sample collected after atomization at 95°F (35°C) will be within the range of 6.5 to 7.2.

(a) Prepare the salt solution at room temperature and adjust the pH so that a 50-ml sample will have a pH between 6.5 and 7.2 after being boiled gently for 30 seconds and cooled to room temperature. The pH of the sample should be measured immediately after it is cooled. It is necessary for the pH of the original solution to be below 6.5 if this method is followed.

(b) Heat the salt solution to boiling and cool to 95°F (35°C) or maintain the solution at 95°F(35°C) for approximately 48 hours before adjusting the pH to the required range of 6.5 to 7.2

(c) Heat the water from which the salt solution is to be prepared to 95°F (35°C) to expel the carbon dioxide before preparing the solution and maintain this temperature while adjusting the pH to within the range of 6.5 to 7.2.

4.9.4.3 Preparation of specimens.

4.9.4.3.1 Specimens. Specimens shall be given a minimum of handling, particularly on the significant surfaces, and shall be prepared for test immediately before exposure. Specimens should not be handled with bare hands.

4.9.4.3.2 Cleaning system. Unless otherwise specified, uncoated metallic specimens and metallic-coated specimens shall be thoroughly cleaned of oil, dirt, and grease as necessary until the surface is free from water break. The cleaning methods shall not include the use of corrosive solvents nor solvents which deposit either corrosive or protective films, nor the use of abrasives other than a paste of pure magnesium oxide. Specimens having an organic coating shall not be solvent cleaned.

4.9.4.3.3 Protected coating. Unless otherwise specified, cut edges and surfaces not required to be tested and the portions of specimens that come in contact with the support shall be protected with a suitable coating of wax or similar substance impervious to moisture.

4.9.4.4 Procedure.



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4.9.4.4.1 Location of specimens. Unless otherwise specified, flat specimens and other specimens where practicable, shall be supported in such a position that the significant surface is approximately 15° from the vertical and parallel to the principal direction of horizontal flow of the fog through the chamber. Other specimens shall be positioned as specified in detail specifications or by the purchaser. Rotating specimen holders may be used provided they meet the requirement of this paragraph except in regard to the position of the specimen with respect to the direction of flow of the fog through the chamber. Whenever practicable, the specimens shall be supported from the bottom or from the side. When specimens are suspended from the top, suspension shall be by means of nylon cord, glass hooks, or waxed string; the use of metal hooks is not permitted. Specimens shall be positioned so that they do not contact each other or any metallic material or any material capable of acting as a wick, so that they are not shielded from the freely settling fog, and so that corrosion products and condensate from one specimen don't fall upon another specimen or return to the solution reservoir.

4.9.4.4.2 Operating conditions.

4.9.4.4.2.1 Temperature. The test shall be conducted with a temperature in the exposure zone maintained at 95°F plus 2°F, minus 3°F (35°C plus 1°C, minus 1.7°C). Satisfactory methods for controlling the temperature accurately are by housing the apparatus in a properly controlled constant-temperature room, by thoroughly insulating the apparatus and preheating the air to the proper temperature prior to atomization, or by jacketing the apparatus and controlling the temperature of the water or of the air used in the jacket. No surface in the chamber shall become overheated enough to cause fog droplets, thereby diluting them. The use of immersion heaters within either (1) the chamber for the purpose of maintaining the temperature within the exposure zone, or (2) in the salt solution reservoir, is prohibited except that suitable heating units may be used in larger exposure chambers provided they do not cause the temperature within the exposure zone to vary outside the permissable range of operating temperature.

4.9.4.4.2.2 Atomization.

4.9.4.4.2.2.1 Exposure zone. The conditions maintained in all parts of the exposure zone shall be such that a suitable receptable placed at any point in the exposure zone will collect from 0.75 to 2.0 ml of solution per hour for each 80 sq cm of horizontal collecting area (10 cm dia) based on an average run of at least 16 hours. When the 5 percent solution is used, the solution thus collected shall have a sodium-chloride content of from 4 to 6 percent [sp. gr. of from 1.0268 to 1.0413 when measured at a temperature between 93°and 97°F(34°and 36°C)] and a pH from 6.5 to 7.2. When the 20 percent solution is used, the solution thus collected shall have a sodium-chloride content of from 18 to 22 percent [sp. gr. of from 1.126 to 1.157 when measured at a temperature between 93° and 97°F (34° and 36°C)] and a pH from 6.5 to 7.2. At least two clean fog-collecting receptacles shall be used; one placed as near to any nozzle and one as far from all nozzles as any specimen. Receptacles shall be placed so that they are not shielded from the freely settling fog and so that no drops of solution from specimens or from other sources will be collected.

4.9.4.4.2.2.2 Nozzles. Nozzles shall be made of material nonreactive to the salt solution. Glass or methyl-methacrylate (Lucite) nozzles with nylon filters are suggested. Suitable atomization has been obtained in boxes having a volume of less than 12 cubic feet with the following conditions:



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(a) Nozzle pressure of 12 to 16 pounds per square inch.

(b) Orifices of from 0.02 to 0.03 inch in diameter.

(c) Atomization of approximately three quarts of the salt solution per 10 cubic feet of box volume per 24 hours.

4.9.4.4.2.2.3 Modified conditions. When using boxes having a volume considerably in excess of 12 cubic feet, the above conditions may have to be modified in order to meet the requirements for operating conditions.

4.9.4.4.3 Length of test. The length of the salt fog test shall be as required by the detail specification or drawing. Unless otherwise specified, the test shall be run continuously for the time indicated or until definite indication of failure is observed, with interruption only for adjustment of the apparatus and inspection of the specimen. Operations shall be scheduled so that interruptions are held to a minimum. The following exposure periods are convenient for scheduling tests and are recommended for use in detail specification: 2, 4, 6, 8, 16, 24, 32, 48, 72, 96, 200, 240, 500, or 720 hours. When a retest is authorized for a sample or samples that failed to pass a specified test, the retest shall be run for the required number of hours with a minimum of interruptions or opening of the cabinet.

4.9.4.4.4 Examination of specimens. At the end of the test, the specimens shall be examined immediately to determine compliance with the standard of acceptability specified in the detail specification or drawing. If necessary to aid in examination, a gentle wash or dip in clean running water not warmer than 100°F (38°C) and blowing with clean dry air may be used. Examination at stated intervals during the test may be required.

4.9.4.5 Report of results.

4.9.4.5.1 Material test. The following information shall be recorded, unless otherwise prescribed in the detail specification or drawing for the material or product being tested.

(a) Solution concentration

(b) Type of salt and water used in preparing the salt solution.

(c) All reading of temperature within the exposure zone of the chamber.

(d) Record of daily test made on each fog collecting device including the following:

(1) Volume of salt solution collected in milliliters per hour per 80 sq cm of horizontal collecting area.

(2) Sodium-chloride content of specific gravity (including temperature at which sp. gr. measurement was made) of the collected solution.

(3) pH of collected solution.

(e) Type of specimen and its dimensions, or number or description of product or part tested.



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(f) Method of supporting specimen in the salt fog chamber.

(g) Description of protection used on cut edges and other surfaces not required to be tested.

(h) Length of exposure period.

(i) Cause and length of interruptions during test.

(j) Results of all inspections.

(k) Method of specimen surface preparation.

## 5. PACKAGING

5.1 Preservation. Preservation shall be Level A, C, or Industrial IAW MIL-STD-2073/1, as specified (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Indicators shall be cleaned in accordance with process IC-I of MIL-P-116.

5.1.1.2 Drying. Indicators shall be dried in accordance with process IC-I of MIL-P-116.

5.1.1.3 Preservative application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-I of MIL-P-116. Each indicator shall be placed in a PPP-B-636 fiberboard container, weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The industrial preservation of the Indicator shall be in accordance with ASTM D3951.

5.2. Packing. Packing shall be Level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare, and cube consistent with the protection required. MIL-STD-2073/1 applies for Army use.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in



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shipping containers conforming to PPP-B-636, Class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use.

5.2.4 Industrial. The packaged Indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tachometer indicator covered by this specification is intended for use to indicate the rotational speed of a jet or prop-jet engine in percent rpm. The calibration is based on the generator's turning 4,200 rpm at 100 percent rated engine speed. The ERU-5A/A indicator is intended for use in aircraft systems requiring a high efficiency anti-reflective coating on the indicator cover glass. The ERU-5/A indicator is intended for use in aircraft systems that do not require a high-efficiency coating on the indicator cover glass.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.

(b) Filling medium (see 3.7.3).

(c) Dial finish requirements if other than as specified in 3.9.1.

(d) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1).

(e) Whether sampling plan B is to be omitted (see 4.4.2.2).

(f) Levels of packaging and packing (see 5.1 and 5.2).

(g) Reinspection date requirements (see 5.3.1).

(h) Type of indication required (see 1.2).

6.3 Definitions

6.3.1 Tapping. The phrase "gently tapped" is defined as meaning the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.3.2 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets.



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6.3.3 Amplitude. The word "amplitude" means the extent of motion as measured from one extreme to the opposite extreme.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the OC-ALC/MMEOR (Code 71), Tinker Air Force Base, OK 73145-5990, and information pertaining to qualification of products may be obtained from that activity.

6.5 Subject term (key word) listing.

Clamp Mounted
ERU-5/A
ERU-5A/A
Hermetically

6.6 International standardization. Certain provisions (3.9.1) (figure 1) of this specification are the subject of international standardization agreement ASCC AIR STD 10/36 and STANAG 3691. When amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:
AIR FORCE -99
Navy -AS

Preparing Activity:
AIR FORCE -71

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER MIL-I-25863D | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|

3. DOCUMENT TITLE
INDICATOR, ELECTRICAL, TACHOMETER, 0-110 PERCENT RPM, 2 INCH

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial (2) AUTOVON |
|---|---|
| c. ADDRESS (Include Zip Code) | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

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